

CS482/495/496 Software Project Proposal:

add your tentative project title here

your name(s) here

2025-10-15

1 Client Information

By sharing this client information and the rest of this document, you are stating that this client has provided this project as something they want (not something you created and asked if they wanted), and that they are interested in having you complete this project for your capstone.

- Client name:
- Client title:
- Client email address:
- Client employer:
- How you know the client:

2 Project Description

2.1 Overview

[Add a few paragraphs describing your project succinctly. What problem are you trying to solve, what is the purpose of your project? Why does your client want this project?]

2.2 Key Features

[At this point you should have a basic understanding of your client's needs. List out the key features of the software system the client wants you to build.]

2.3 Why this Project is Interesting

[Why did you decide this project was interesting enough to you to be a capstone project? What about this project is enticing? Why should anyone care?]

2.4 Areas of CS required

[What subfields of computer science seem most likely to be relevant to your project? A capstone must involve multiple.]

2.5 Potential Concerns and Questions

[Is there any aspect of this project that makes you unsure if it will work, either due to your own interests/background, or that you aren't sure if it fits the requirements? Are there questions you have about this project that you want instructor feedback about?]

2.6 Summary of Efforts to Find a Project

(Not necessary for 482) [Briefly list out when/how you've discussed with this client, and if you've discussed with other clients who either didn't work out or didn't respond. If you considered a different project and it didn't work out, why didn't it work out?]

[Most CS495 projects end here. The sections below are for CS482 and CS496 software projects].

2.7 Comparison to Draft

[For CS496 only, focus on highlighting the major differences between the draft proposal in CS495 and this one here. If there are no major differences, you can remove this subsection.]

3 Requirements

3.1 Non-Functional Requirements

[Non-functional requirements are just as important as functional requirements. Dont forget to specify them.]

ID	NFR Title	Category	Description
NFR1	NFR Example 1	Usability	Description of the NFR (it does not follow a user story template)
NFR2	NFR Example 2	Security	Description of the NFR (it does not follow a user story template)

Table 1: Non-Functional requirements

3.2 Functional Requirements (User Stories)

[In CS482, all functional requirements are written as User Stories. In CS496, some projects may use a different template to write the requirements. The table below is an example of writing the Stories. Adapt accordingly to different templates or if you want to display more info.]

ID	Story Title	Points	Description
S1	Story Example 1	5	As a user, I want to write a user story example, so that people will understand them.
S2	Story Example 2	2	As a user, I want to write a user story example, so that people will understand them.

Table 2: Functional requirements as User Stories.

4 System Design

4.1 Architecture

For this project, we will be using the MVC architecture. This architecture will be composed of three layers, each with their own modules.

The Model layer defines data schemas and handles database communication with MongoDB collections via Mongoose. Including UserModel to manage user data such as name, email, and password, EventModel to store calendar event details like title date, time, description. An AuthModel to manage authentication tokens and session validation and a Data Base connection model to connect to the production and test mongo database.

The View layer consists of front-end templates and React components including the Login/Register page, the main Dashboard, EventForm, CaladarView, and ProfilePage

The Controller layer receives user requests, manipulates the Model, and returns a View or data Which could include the UserController, AuthController and EventController

4.2 Diagrams

[CS482, on sprints/iterations 2-3, you need to create and update a diagram (check the assignment for which type of diagram). On CS496, since before sprint/iteration 1 you should have a class diagram and keep it up-to-date.]

4.3 Technology

For our frontend, we will be using the React framework, and we will be using Vite as our build tool. We will also be using Tailwind Css for our styling. For our backend, we will be using Node.js for our runtime, Express.js for our framework, bcrypt for our password hashing, and JSON Web Token for our authentication. For our database, we will be using MongoDB with Mongoose for schema modeling and database interaction. And for the calendar we will use FullCalendar, an open source JavaScript library for event management and displaying the calendar. We will be using Jest for our testing, along with the build-in mock system inside of Jest. We will be using dotenv to securely load our environment. For our deployment, we will be using Railway.

4.4 Coding Standards

We will be using Pascal case for our schemas and file names, and all lowercase for our folder names. For our variable and function names in JS, we will be using Camel case. For our design, we will focus on simplicity, functionality, and ease of use over pure looks.

All code that is pushed to main should have unit tests and coverage that is over 70%. We will all use the project board to communicate which tasks we are working on. When working on the project, we will create our own branches which will only be merged with the main branch when they have been tested and reviewed. We will implement a .env file to hide sensitive information. We will follow our user stories and always include the # in our commits so they link to the user story we are working on. We will commit frequently and always include meaningful commit messages. When we find a bug, we will add tests to specifically test that the bug has been eliminated, so we know if that bug ever shows up again. When testing our code, we will use our test database and not the main database. Our code will be object-oriented and modular to improve readability. And if needed we can implement pair programming to tackle big/ complex user stories.

4.5 Data

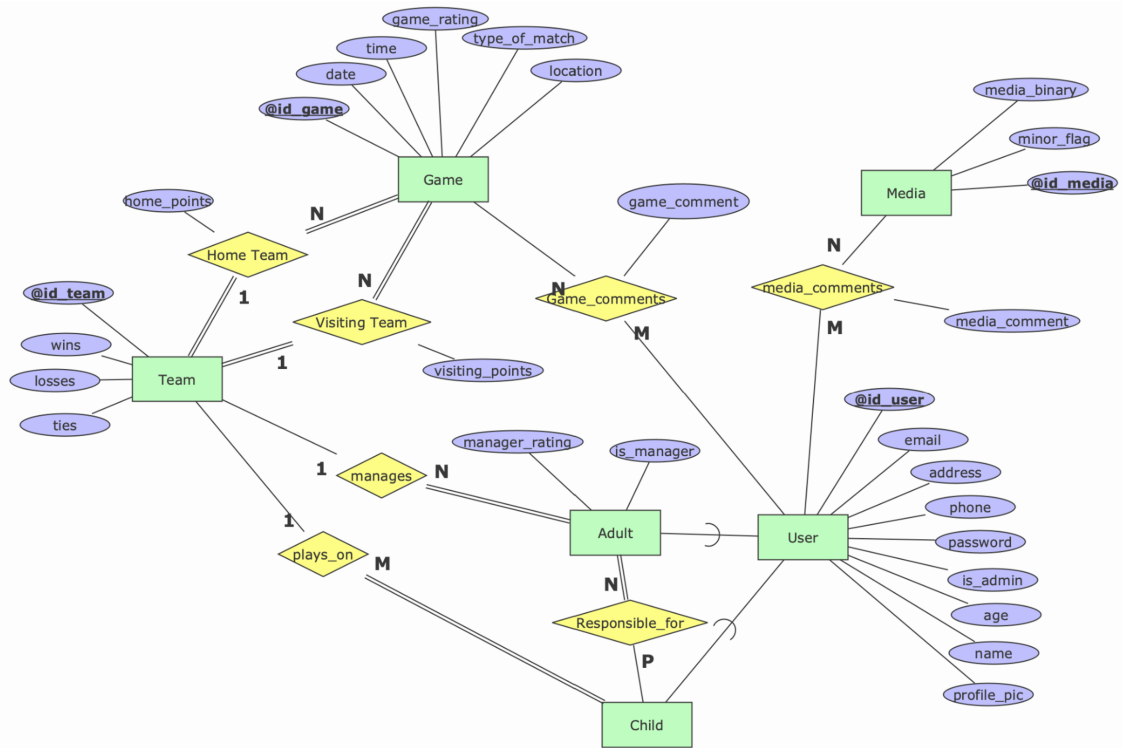


Figure 1: Our ERD

4.6 UI Mocks

edit Profile

 change profile pic

email	change email
phone	confirm email
address	submit
age	change password
name	confirm password
	current password
	submit

your children:
Bill, Jimbo

child email
child password
add child

Figure 2: A page to edit your profile

Past games

teams	score	date	location	rating	match type
- - -	..	.	.	...	- - -

Figure 3: A page to see all past games

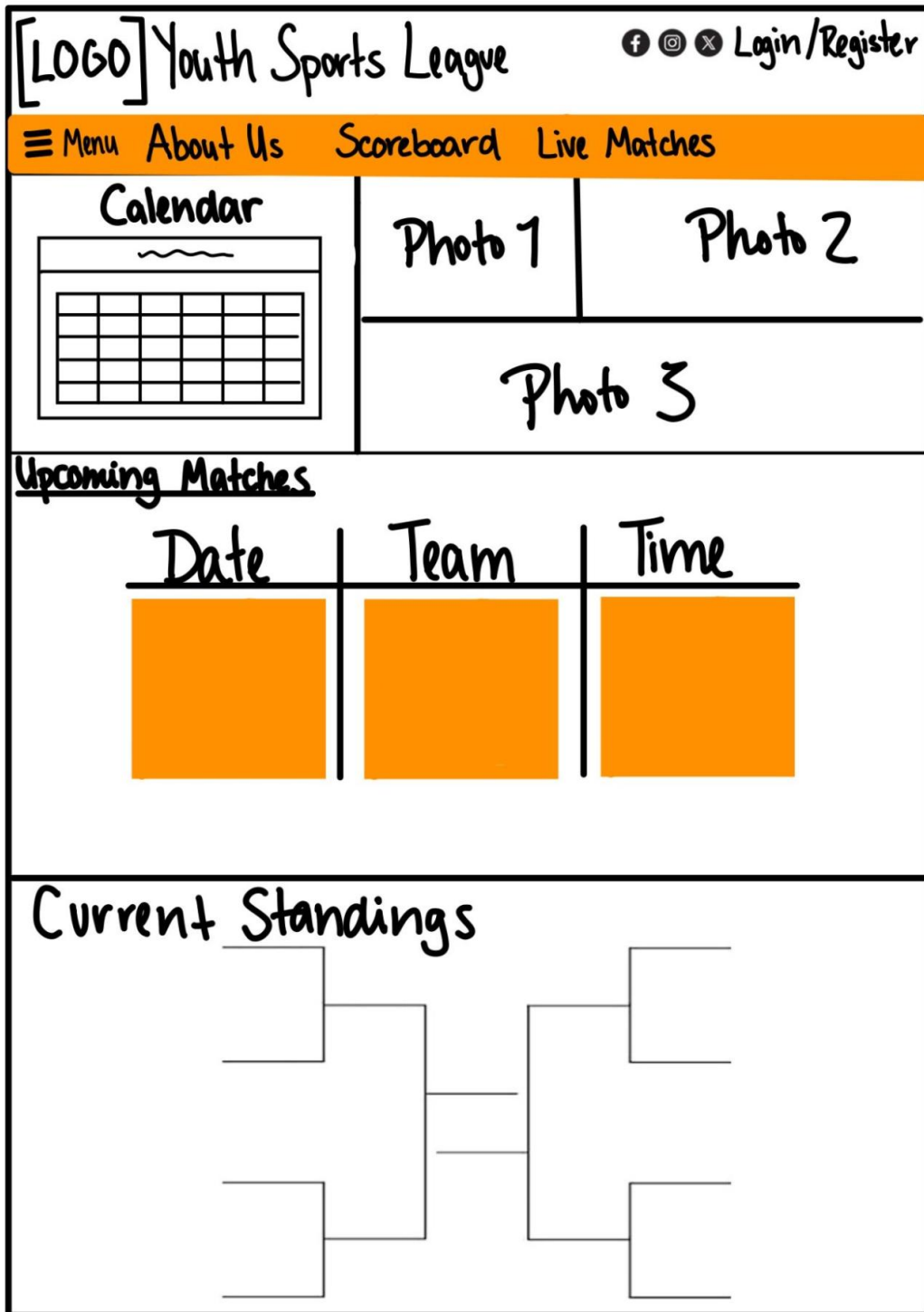


Figure 4: The main landing page with calendar, bracket, user posted images, and upcoming games

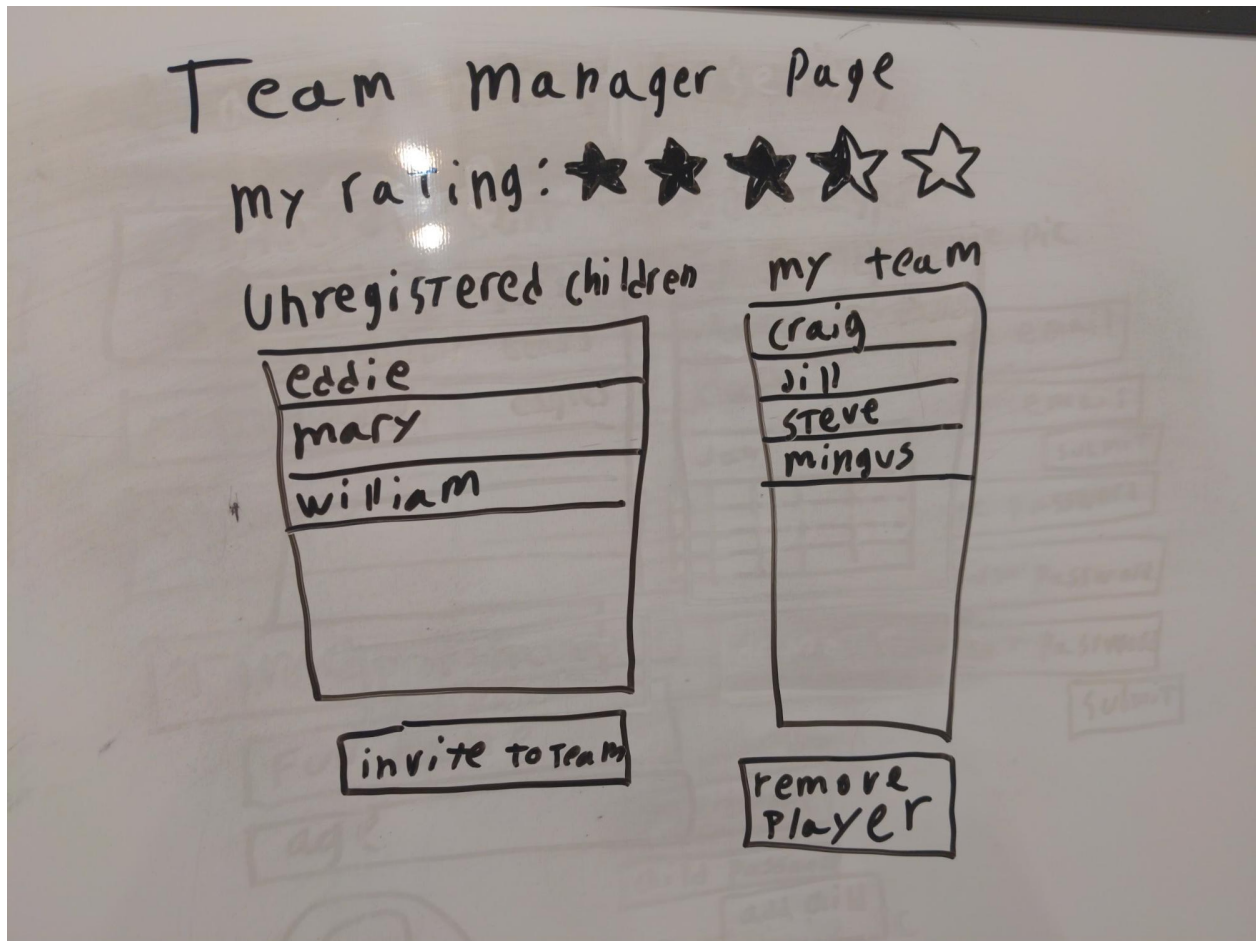


Figure 5: A page for the team manager to see their score and manage their team

YOUR TEAM :

PROFILE REMOVE	PROFILE REMOVE	PROFILE REMOVE	PROFILE REMOVE
PROFILE REMOVE	PROFILE REMOVE	PROFILE REMOVE	PROFILE REMOVE
PROFILE REMOVE	PROFILE REMOVE	PROFILE REMOVE	PROFILE REMOVE
PROFILE REMOVE	PROFILE REMOVE	PROFILE REMOVE	PROFILE REMOVE

ADD PLAYERS :

PROFILE ADD	PROFILE ADD	PROFILE ADD	PROFILE ADD	PROFILE ADD
PROFILE ADD	PROFILE ADD	PROFILE ADD	PROFILE ADD	PROFILE ADD

Figure 6: An alternative team manager page

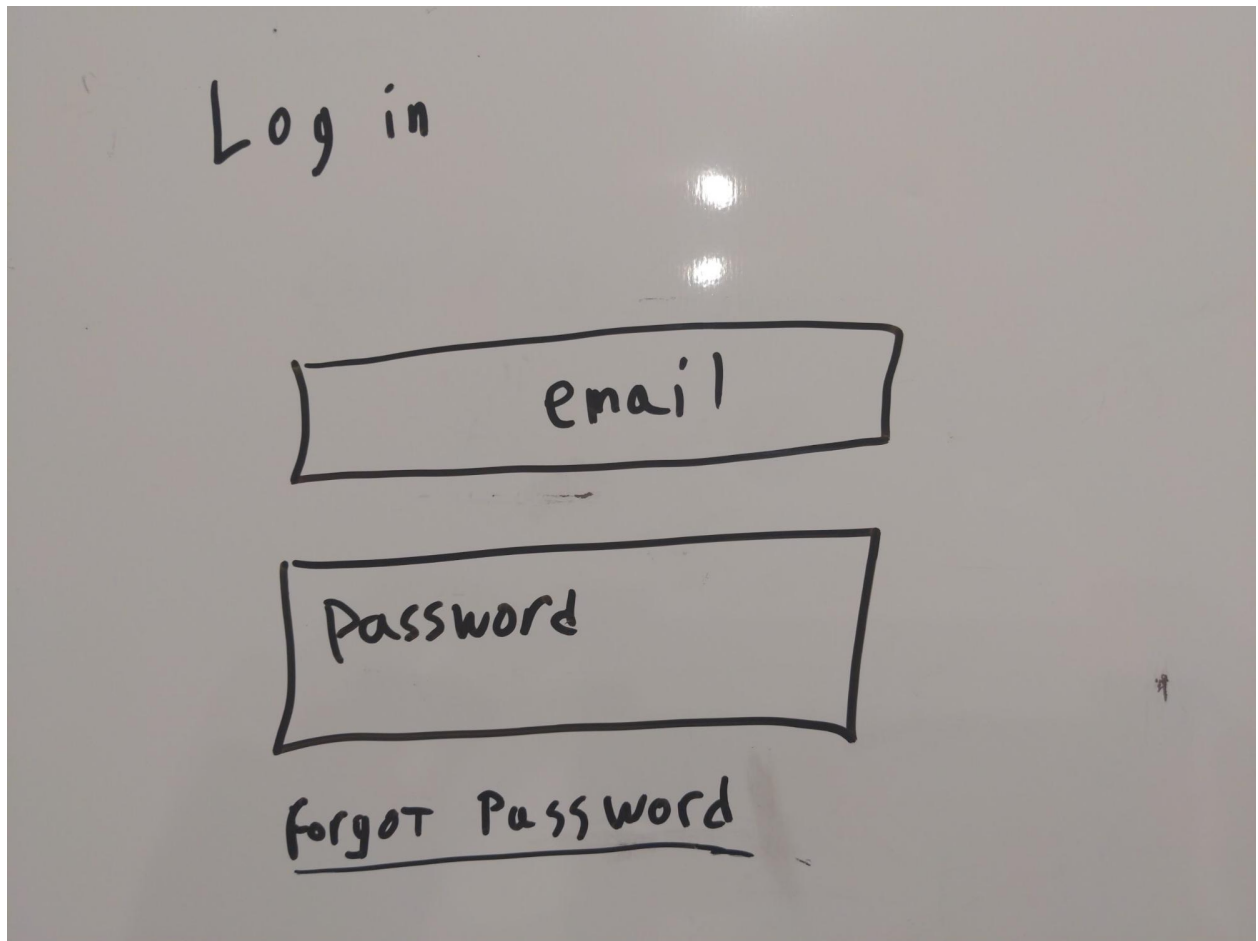


Figure 7: A basic login page

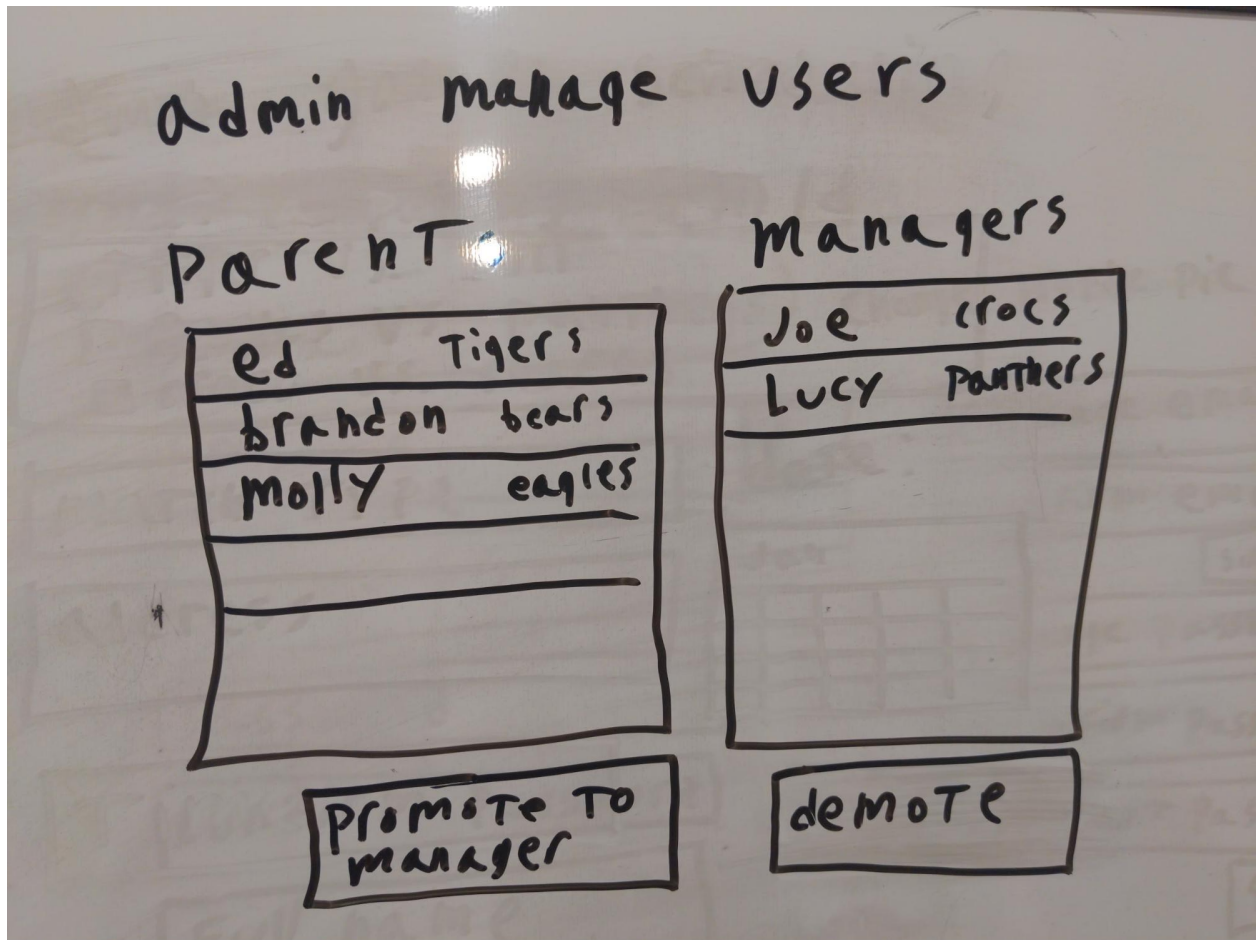


Figure 8: A page for admins to promote and demote team managers

view all league media

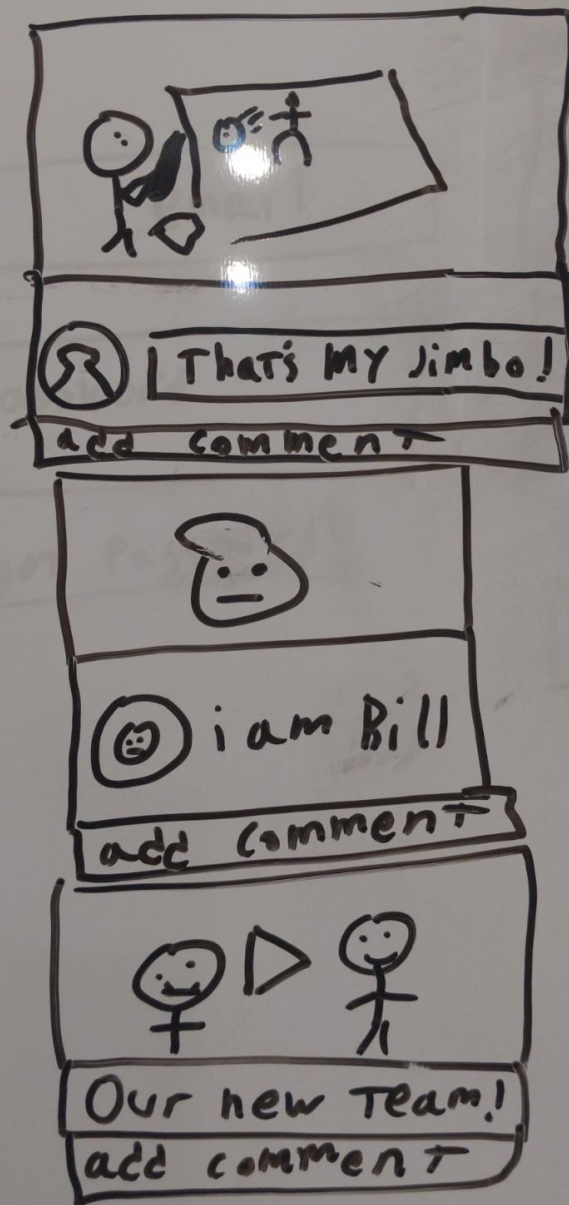


Figure 9: A page to see all previous posts by users and comment on their posts

admin game scheduling

select a pairing

☐ Tiger vs. bull
☐ eagles vs. panthers
☒ crocs vs. sharks

match TYPE

address

Time

date:

Jan				

Figure 10: A page for admins to schedule the games that still need to happen for the season

Create Profile

parent ☐ child

email

phone

address

password

confirm password

Full name

age

 + add profile pic

Figure 11: A basic account creation page

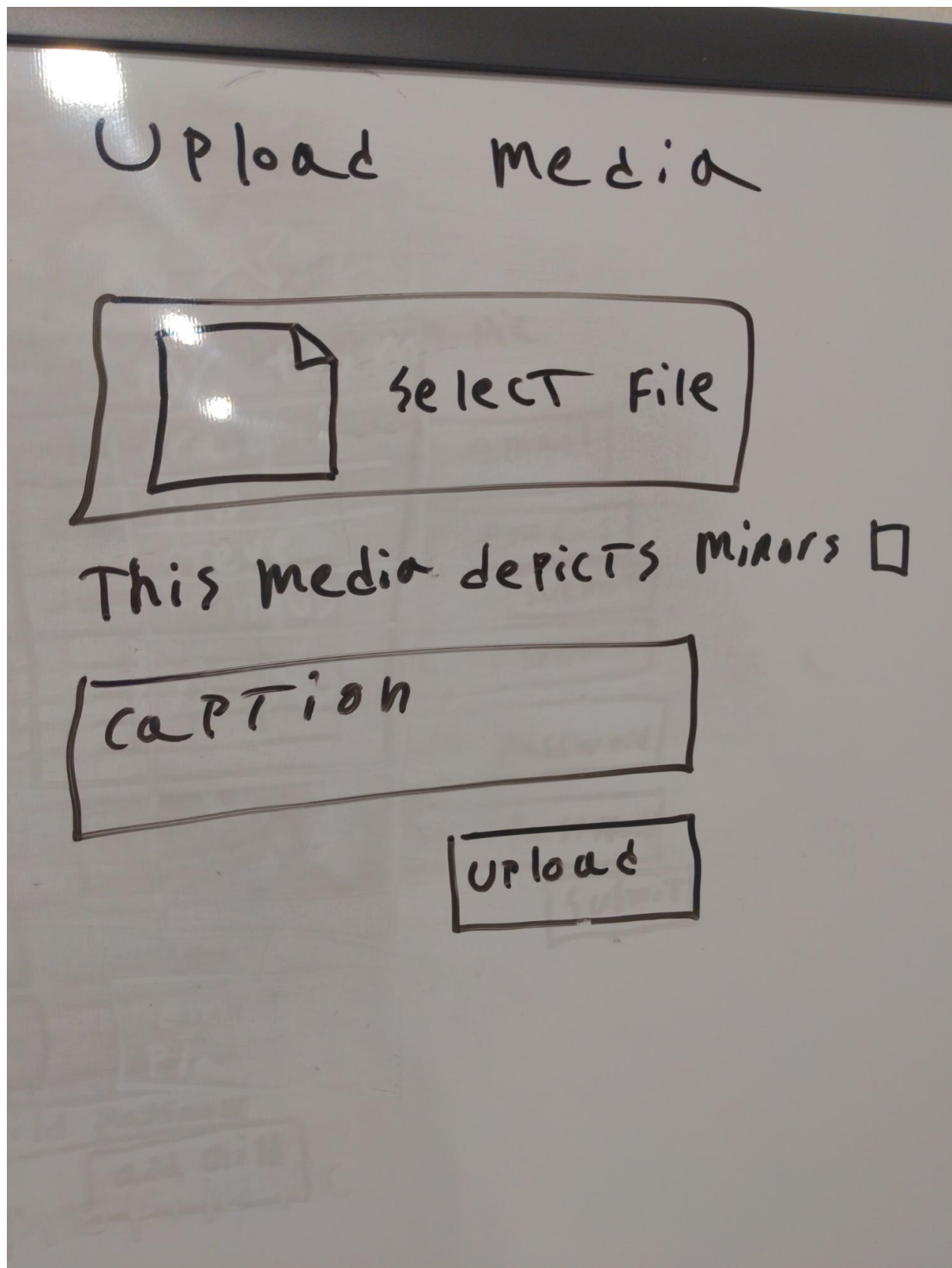


Figure 12: An upload page for uploading pictures and videos of the league

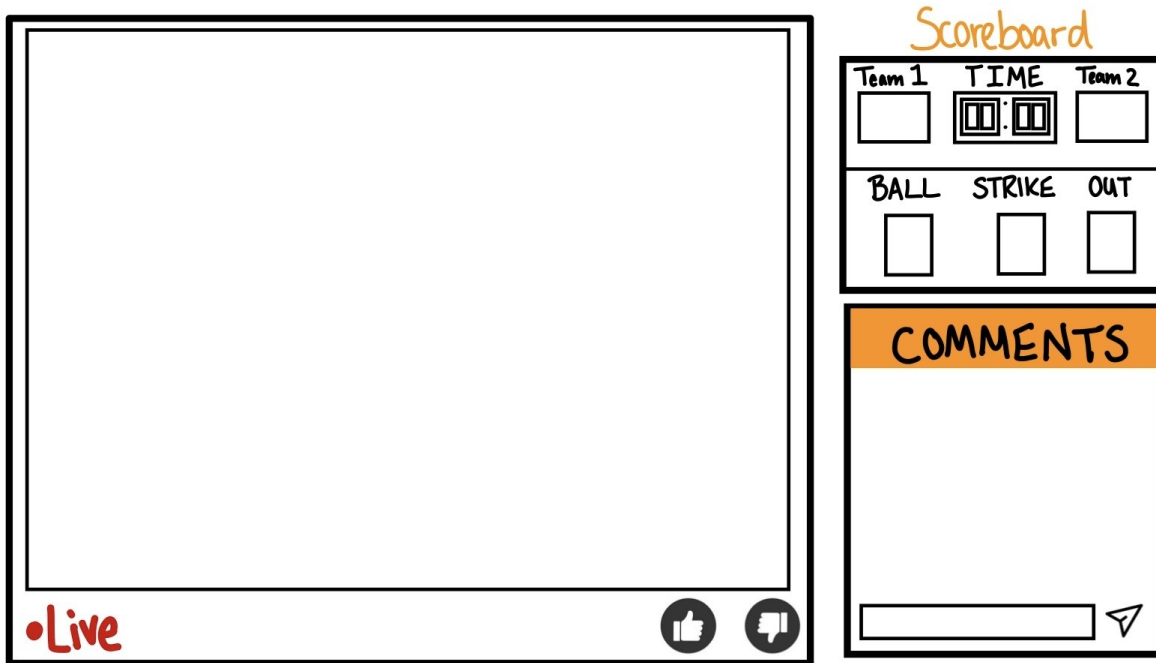


Figure 13: A page to see the current game score, rate the game, and comment your thoughts

5 Iterations

5.1 Iteration Planning

[In CS496, you plan all iterations beforehand. In CS482, you update the planning here at each iteration.]

Iteration	Dates	Stories	Points
1	01/01 - 02/01	S1 Story Example, S2 Story Example 2	07
2	02/01 - 03/01	S3 Story Title, S4 Story Title, S5 Story Title, S6 Story Title	17
3	03/01 - 04/01	S7 Story Title, S8 Story Title, S9 Story Title, S10 Story Title, S11 Story Title	21
4	04/01 - 05/01	S12 Story Title, S13 Story Title, S14 Story Title, S15 Story Title	19
5	05/01 - 06/01	S16 Story Title, S17 Story Title	06
Total:			70

Table 3: Iteration Planning for Incremental Deliveries

5.2 Iteration/Sprint 1

5.2.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by

completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.2.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.2.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.2.4 Retropective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.3 Iteration/Sprint 2

5.3.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.3.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.3.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.3.4 Retropective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.4 Iteration/Sprint 3

5.4.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.4.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.4.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.4.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.5 Iteration/Sprint 4

[CS496 has 5 sprints. CS482 only has only 3 sprints (remove Iterations 4 and 5 from this doc if you are writing a doc for 482)]

5.5.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.5.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.5.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.5.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.6 Iteration/Sprint 5

5.6.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.6.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.6.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.6.4 Retroerspective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

6 Final Remarks

6.1 Overall Progress

[Have you completed everything? If so, present evidence on how you brought value to your client, and the overall client satisfaction. Otherwise, estimate how much progress you done and how long it would take to finish this project.]

6.2 Project Reflection

[Your personal reflection on the project. What lessons did you learned. What would you have done differently. How can you do better work in future projects? You may write this as a team or per person (or both)]

Appendix

[Appendix section if needed]